

Made Arbi Parameswara

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PROFESSIONAL SUMMARY

Proven expertise in designing reusable **Python scraper libraries**, **Docker and Kubernetes orchestration**, and **CI/CD pipelines (GitHub Actions, Helm)**, delivering reliable **daily financial data ingestion** (IDX, OJK, BI, news) monitored by Prometheus/Grafana monitoring. **Demonstrated leadership as Assistant Team Lead**, mentoring juniors, defining coding standards, and **optimizing cloud costs (GKE savings: \$1.8K/year, MLflow 90x latency reduction)**. Skilled in Medallion Architecture ETL/ELT and stakeholder collaboration to productionize scraper outputs for ML pipelines and financial analytics. C2 English proficient (EF SET 76/100).

WORK EXPERIENCES

Data Engineer (Machine Learning Engineer) (Contract) | Mandiri Sekuritas Nov 2025 – Present

- Supporting implementation of **GCP-native Data Platform with Airflow (Cloud Composer), BigQuery, and Cloud Storage**, following **Medallion Architecture** for automated daily ingestion of **IDX, BI, and OJK financial data**.
- Developing Selenium-based news scrapers to enrich datasets from **financial news from Indonesian sources, integrated into ETL pipelines**.
- Designing and implementing a centralized Machine Learning Platform on Kubernetes (GKE)** for streamlined training, deployment (ML Model), monitoring, and governance (AI & ML Models) at Mandiri Sekuritas using **Kubeflow** (orchestration), **Feast** (feature store), **BentoML** (model serving), **Evidently** (Data drift detection), **MLflow**, and **Data Hub** (Data Governance) **saving ~Rp 500M (USD 30K)/year and improving model reability**.
- Optimized MLflow logging** with 90x latency reduction (3.71s → 0.04s (median)) for agentic workflows and inference via direct PostgreSQL backdoor logging bypassing API overhead and **cut CPU usage 22-23%, memory accumulation ~774MB, yielding \$1,440-1,800/year GKE savings (2 vCPUs, 8GiB RAM reduced)**.

Assistant Team Lead (Contract) | BPJS Kesehatan May – Nov 2025

- Served as an assistant team lead in the BPJS AI Platform project**, acting as a **team leader in the Anti-Fraud team at BPJS Kesehatan** with a technical focus on **fraud detection and risk scoring models**.
- Led the anti-fraud team and implemented the system on BPJS's on-premises server, capable of processing 6+ millions claims/month from unstructured PDFs, multiple templates, and traditional data, including handwritten forms, blurry, and skewed scanned documents from 3,000+ hospitals across Indonesia.
- Set up and monitor the implementation of MLOps using **CI/CD pipelines with SQLServer, Git, GitHub, GHCR.io, Docker, Helm, Rancher, Kubernetes (K8s)**, and monitoring via **Grafana and Prometheus**.
- Implemented inference serving with **containerized microservices with 10+ features ETL and integrated with Apache Airflow** to set up a batch processing system and a streaming processing workflow for demonstration purposes with Apache Airflow and Kafka.
- Oversee and guide experiments, inference tests, and KPI definition for product evaluation.
- Prepare initial technical documents to facilitate easier project documentation and audits by external institutions, such as BPK.

AI & Software Engineer (Contract) | BPJS Kesehatan May – Nov 2025

- Use the latest AI stacks, such as Polars, PyTorch (with GPU acceleration), and ONNX, to optimize and enhance concurrency, making processes **10x faster compared to the initial design**.
- Train and optimize models using hyperparameter tuning on classifiers, using MLP (IndoBERT + ResNet50) in FP16 and YOLOv8s (handwritten signature & QR) in ONNX format, achieving an F1-score of ~86%.
- Optimize Qwen-2.5-VL-7B-instruct with streaming off and precision FP16 vLLM inference (text + image) for extraction, achieving user-acceptable extraction accuracy of ~90% on test data from unstructured PDFs and ~70% on handwritten PDFs.
- Create an explainable risk assessment system for flagged claims using Qwen-2.5-VL-7B-instruct with streaming off and Precision FP16 vLLM inference (text only), supporting verificatory decision with an initial user acceptance rate of ~90%.
- Linked the BPJS Kesehatan **SQL Server database (RDBMS)** to the system for data storage and input and utilize on-premises S3-minio object storage to save supporting data for the claims.
- Develop a login feature that integrates **Redis** for session management, **JWT** for authentication, and connects to the **SQL Server database** at BPJS Kesehatan to ensure seamless and efficient user authentication and session handling.

Lead Software Engineer (Contract) PT. Gravicode Multinovative Plexindo	March – May 2025
- Led a development of a finance app with 30+ features using C#, JavaScript, .NET 8, ASP.NET Core, Blazor, and SQL Server, successfully creating scalable and maintainable solutions. - Managed automatic pull requests (PR) in GitHub Actions to ensure continuous integration (CI) and high-quality code. - Collaborated with cross-functional teams (DevOps, QA, Project Manager) to ensure technical quality of the system, seamless integration, automated testing, and deployment in CI/CD pipelines. - Employed microservices architecture and RESTful API development to streamline finance management modules. - Utilized Entity Framework ORM for database connections with Microsoft SQL Server and implemented code-first database approach for better maintainability and scalability. - The project was postponed due to a default on the client's end.	
Machine Learning Engineer (Internship) PT. Jasamedika Saranatama	May – Aug 2023
- Designed text extraction model using Pytorch, Pandas, and NumPy, achieving a micro F1-score of 0.841, implementing CUDA trained, and performing hyperparameter tuning. - Supporting the development of a content-based filtering recommender system for disease prediction, using LightGBM, KNN (Scikit-Learn), and SHAP, achieving 0.822 accuracy and 0.824 F1-score.	

PROJECTS

Text Extraction BIOGAN-BERT: BioGPT-2 Fine Tuned and GAN-BERT for Extracting Drug Interaction Based on Biomedical Texts	Jan – Aug 2024
Enhanced model performance by incorporating deep learning and novel data augmentation using LLM (BioGPT2) and GAN for text extraction, achieving a micro F1-Score of 85% (State of The Art (SOTA) 2024). Integrated generative AI data augmentations to optimize model generalization and improve prediction accuracy, demonstrating hands-on skills with LLMs and data processing techniques.	
Big Data Automated Streaming Cryptocurrency Data Analysis	Jan – May 2024
- Built a scalable infrastructure using Kafka, Elasticsearch, and Spark to process high-velocity data and enable real-time cryptocurrency price prediction and sentiment analysis (< 1-minute latency). Achieved a grade of "A" (4.00/4.00) for the project, delivering a high-performance, low-latency system under the supervision of the CTO of NoLimit.id.	
Vision Rice Disease Image Classification (Vision)	Sept – Dec 2022
- Designed rice disease classification model, achieving perfect (1.00) Accuracy and F1-Score performance and recognized as a finalist and a favorite team in TSDN 2022. - Developed the system with REST APIs and a CI/CD pipeline, utilizing Git, Flask, Docker, and AWS. - Utilized Pandas and NumPy for data preparation and created visualizations with Matplotlib and Seaborn to communicate data and model performance.	

EDUCATION

Bandung Institute of Technology (ITB)	2022 - 2024
M.Sc. in Informatics, CGPA: 3.54/4.00	
Specialization : Data Science & Artificial Intelligence Notable courses : Big Data, Intelligence Business Analytic, Artificial Intelligence, Deep Learning	

PUBLICATIONS

- ICACTA (2024): BIOGAN-BERT: BioGPT-2 Fine Tuned and GAN-BERT for Extracting Drug Interaction Based on Biomedical Texts. - ICEEI (2023): Big Five Personality Prediction Based on Indonesian Tweets and Personality Test (NLP).	
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QUALIFICATIONS

EF SET English Certificate 76/100 (C2 Proficient)	2025
Finalist & Favorite Team in Turnamen Sains Data Nasional	2022
Machine Learning with Python Cognitive Class – IBM	2022
International English Language Testing System (IELTS): Average Score of 7 British Council	2022
Elite Executive Workshop - Lifetime Program (Level 1 - 5)	2020